ETALE DESCENT DATA ALONG A PROPER SURJECTIVE MORPHISM ARE EFFECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 965-974 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 976-994, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 30 of queue_spaces-pushouts.json (pb=972, stmt 965-974, proof 976-994).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GFW: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 817–828. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ADG: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 994–994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ADQ: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 994–994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GF6: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 994–994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03EB: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 994–994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a proper surjective morphism of algebraic spaces over S. Any descent datum $(U/X, \varphi)$ relative to f (Descent on Spaces, Definition 0ADG) with U étale over X is effective (Descent on Spaces, Definition 0ADQ). More precisely, there exists an étale morphism $V \to Y$ of algebraic spaces whose corresponding canonical descent datum is isomorphic to $(U/X, \varphi)$.*

Proof. Recall that U gives rise to a representable sheaf $\mathcal{F} = h_U$ in $Sh(X_{spaces,étale}) = Sh(X_{étale})$, see Properties of Spaces, Section 03EB. The descent datum on U relative to f exactly gives a descent datum $(\mathcal{F}, \varphi)$ for étale sheaves with respect to $\{X \to Y\}$. By Lemma 0GFW this descent datum is effective. Let $\mathcal{G}$ be the corresponding sheaf on $Y_{étale}$. By Properties of Spaces, Lemma 0GF6 we obtain an étale morphism $V \to Y$ of algebraic spaces corresponding to $\mathcal{G}$; we omit the verification of the set theoretic condition[1]. The given isomorphism $\mathcal{F} \to f_{small}^{-1}\mathcal{G}$ corresponds to an isomorphism $U \to V \times_Y X$ compatible with the descent datum. □

[1]It follows from the fact that $\mathcal{F}$ satisfies the corresponding condition.